

SANGAR, Russian Federation

WMO: 246520

Lat: 63.959N

Lon: 127.421E

Elev: 96

StdP: 100.18

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.1	-46.5	-50.0	0.0	-48.0	-48.4	0.0	-46.4	10.2	-26.9	8.5	-28.8	0.6	300	0.761

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	28.0	18.6	26.2	17.8	24.3	16.9	20.0	25.5	18.9	24.1	17.8	22.8	3.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.1	22.4	16.9	12.2	21.8	15.7	11.3	20.6	58.1	25.4	54.3	24.1	50.9	22.8	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.4	7.3	-49.3	31.0	2.0	1.8	-50.7	32.3	-51.8	33.4	-52.9	34.4	-54.3	35.6		
(5)			-49.3	22.2	2.0	1.3	-50.7	23.1	-51.8	23.9	-52.9	24.7	-54.3	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.9	-37.3	-32.9	-20.0	-5.9	5.5	15.1	18.4	14.8	5.4	-8.0	-27.0	-36.7	(6)
(7)		DBStd	21.02	5.80	5.80	7.45	6.08	4.90	4.37	3.79	3.71	4.19	6.20	7.90	6.23	(7)
(8)		HDD10.0	7510	1466	1202	931	477	157	9	1	5	146	558	1110	1449	(8)
(9)		HDD18.3	10028	1724	1436	1189	727	399	113	48	122	387	816	1360	1708	(9)
(10)		CDD10.0	599	0	0	0	0	16	162	260	153	9	0	0	0	(10)
(11)		CDD18.3	77	0	0	0	0	0	17	49	11	0	0	0	0	(11)
(12)		CDH23.3	602	0	0	0	0	0	151	386	66	0	0	0	0	(12)
(13)		CDH26.7	122	0	0	0	0	0	24	89	8	0	0	0	0	(13)
(14)	Wind	WSAvg	3.1	2.2	2.3	3.0	3.6	3.8	3.6	3.3	3.2	3.4	3.5	2.9	2.2	(14)
(15)	Precipitation	PrecAvg	337	11	10	12	15	28	40	51	50	45	42	20	12	(15)
(16)		PrecMax	473	26	34	38	42	65	132	126	116	120	86	51	31	(16)
(17)		PrecMin	198	2	0	1	2	0	6	10	6	0	8	6	3	(17)
(18)		PrecStd	65	7	8	9	8	15	26	30	25	27	19	11	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-20.7	-15.1	0.5	10.0	21.1	29.2	31.0	27.7	19.4	4.7	-7.0	-19.8	(19)
(20)		MCWB	-21.1	-15.7	-2.1	4.7	10.8	18.1	19.7	19.3	12.9	2.1	-8.0	-20.2	(20)	
(21)		2%	DB	-23.7	-17.7	-3.0	7.3	18.5	27.0	29.0	25.0	16.0	2.2	-10.2	-22.7	(21)
(22)		MCWB	-24.1	-18.2	-4.8	2.5	9.8	17.1	19.0	17.7	11.1	0.4	-10.8	-23.0	(22)	
(23)		5%	DB	-26.3	-20.9	-5.8	5.0	16.0	25.0	27.2	23.3	14.0	0.8	-12.6	-24.8	(23)
(24)		MCWB	-26.5	-21.3	-7.4	1.1	8.7	15.9	18.7	16.6	9.9	-0.4	-13.1	-25.1	(24)	
(25)		10%	DB	-28.7	-23.8	-8.7	2.9	13.6	23.0	25.5	21.4	11.9	-0.5	-15.0	-27.4	(25)
(26)		MCWB	-28.9	-24.1	-10.0	-0.4	7.4	15.0	18.0	15.7	8.6	-1.5	-15.5	-27.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-21.0	-15.6	-2.1	4.7	11.8	20.2	22.0	20.3	13.7	2.8	-7.8	-20.2	(27)
(28)		MCDWB	-20.7	-15.2	0.3	9.5	18.2	27.0	28.1	25.7	17.7	4.3	-6.9	-19.8	(28)	
(29)		2%	WB	-24.0	-18.2	-4.8	2.8	10.5	18.2	20.6	18.7	12.1	0.7	-10.7	-22.9	(29)
(30)		MCDWB	-23.7	-17.7	-3.0	6.6	17.2	24.5	26.6	23.4	14.8	1.7	-10.2	-22.7	(30)	
(31)		5%	WB	-26.5	-21.3	-7.4	1.3	9.2	17.0	19.7	17.5	10.5	-0.3	-13.1	-25.0	(31)
(32)		MCDWB	-26.3	-20.9	-5.9	4.4	14.9	23.0	24.9	21.7	13.4	0.7	-12.6	-24.8	(32)	
(33)		10%	WB	-28.9	-24.1	-10.0	0.0	7.9	16.0	18.8	16.3	9.1	-1.6	-15.4	-27.6	(33)
(34)		MCDWB	-28.7	-23.8	-8.7	2.7	13.0	21.7	23.7	20.4	11.5	-0.5	-15.0	-27.4	(34)	
(35)	Mean Daily Temperature Range	MDBR	5.5	6.8	9.7	10.4	8.7	9.6	9.3	8.7	7.2	5.0	5.7	5.0	(35)	
(36)		5% DB	MCDBR	8.0	7.8	10.6	10.2	11.4	13.0	12.0	11.3	10.8	5.2	6.6	6.9	(36)
(37)		MCWBR	7.9	7.5	9.0	6.9	6.1	6.2	5.6	5.8	6.5	4.0	6.3	6.7	(37)	
(38)		5% WB	MCDBR	8.0	7.7	10.3	9.3	10.4	11.0	10.4	9.7	9.4	4.9	6.6	6.9	(38)
(39)		MCWBR	7.9	7.4	8.8	6.4	5.8	6.0	5.5	5.4	6.0	4.0	6.2	6.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.193	0.248	0.293	0.345	0.399	0.414	0.492	0.472	0.344	0.270	0.193	0.180	(40)	
(41)		taud	1.910	2.077	2.077	2.072	2.191	2.265	2.026	2.027	2.381	2.266	1.921	1.884	(41)	
(42)		Ebn at Noon	382	668	781	818	806	802	711	679	737	644	364	163	(42)	
(43)		Edh at Noon	29	64	99	126	124	118	147	134	76	52	27	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.21	1.10	2.92	4.99	5.36	5.95	5.40	4.00	2.38	0.94	0.35	0.08	(44)	
(45)		RadStd	0.01	0.05	0.12	0.21	0.47	0.40	0.27	0.34	0.33	0.12	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	-162	N/A	N/A

Nomenclature: See separate page